25901146 ***SCHEME***

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-BBA(HONOURS) Regular Examinations March 2026
Fourth Semester
VAC-MG4VACBBA200 - MANAGEMENT INFORMATION SYSTEMS AND APPLIED DATA ANALYSIS
(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes *Maximum Marks: 50*

Remember (k), Understand (U), Apply(A), Analysis (An), Evaluate (E), Create (C) , Skill (S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A MCQ
Answer all Questions
Each question carries 1 mark

1.Edge computing processes data:
c) Closer to the data source
2. ERP and CRM systems are examples of:
b) Enterprise applications
3. The set of rules that governs the communication between different computers on a network is called a:
b) Protocol
4. A legacy organizational culture that resists transparency is creating data silos despite the implementation of a state-of-the-art ERP system. Which dimension of the information system is primarily failing to align?
a) The Organization Dimension
5. Which level of management uses MIS the most?
c) Middle level only
6. Mainframe computing is characterized by:
b) Centralized processing
7. IT infrastructure evolution began with:
b) Mainframe systems
8. Inventory control in E-Business is an example of:
a) Internal electronic process
9. Which report is generated at fixed intervals?
b) Scheduled report
10. E-communication involves:
a) Email
11. Forward chaining answers the question:
a) Why did this happen?

12. Tagging in enterprise social networking helps to:
b) Add keywords to identify content
13. MIS reports are generally:

a) Periodic
14. Pearson's correlation measures:
a) Linear relations between variables
15. What is the assumption of ANOVA?
d) All
16. Number of books borrowed by students from a library in a month is an example of:
b) Discrete quantitative data
17. In simple linear regression there are:
a) One independent and one dependent variable
18. Which of the following is an example of qualitative data?
c) Eye Color
19. Logistic regression is used for
b) Classification
20. Where do we get the option to customize the variable in Jamovi?
a) Variable panel

Part B – Short Essay
Answer any 6 questions
Each question carries 5 Marks

21. Discuss how MIS improves organizational efficiency. (any 4 points with correct explanation give 5 marks)

Management Information System (MIS) improves organizational efficiency by:
Providing accurate and timely information.
Supporting managerial decision-making.
Reducing paperwork and duplication of work.
Improving communication and coordination among departments.
Enhancing productivity through automation.
Helping in planning, monitoring, and controlling business activities.

22. What is the difference between Information and Knowledge?
(any 3 points with correct explanation give 5 marks)

Information	**Knowledge**
Processed data with meaning	Information combined with experience and understanding . it is an asset.
Answers "what", "where", "when"	Answers "how" and "why"
Easy to store and transfer	Difficult to transfer. It has different forms(explicit \ itactic).

23. What is the priority in Business Process Re-engineering (BPR)?(Give 5 marks to the correct description)
The primary priority of BPR is to achieve dramatic improvements in business performance by redesigning existing processes.
Objectives:

- Reduce cost
- Improve quality
- Increase speed and efficiency
- Improve customer satisfaction
- Eliminate unnecessary activities

24. What are the prerequisites for E-communication?
(Give 5 marks to the correct description)
Requirements for E-communication:
Computer or smart device
Internet connection
Communication software (Email, Chat, Video conferencing)
Network infrastructure
User accounts and security measures

25. Explain Knowledge Management System (KMS).(Give full marks to any correct description)

A Knowledge Management System (KMS) is a system that helps organizations Acquire, store, share, and Apply
(Knowledge Discovery,Analysis,expert knowledge networks,Knowledge Storage etc)
Different types of KMS,
Functions:
K*nowledge creation*
Knowledge storage
Knowledge sharing
Knowledge retrieval
Benefits:
Better decision-making
Increased productivity
Reduced duplication of work
Improved organizational learning

26. Explain the difference between discrete and continuous data with examples.
(any 4 points give 5 marks)

Discrete Data	Continuous Data
Countable values	Measurable values
Takes specific numbers	Can take any value within a range
No fractions usually	Fractions and decimals possible
Example: Number of students	Example: Height, weight, temperature,

27. What are the assumptions of the Wilcoxon Signed-Rank Test (Paired Sample)?
(any 4 points give 5 marks)

- Data are paired observations.
- Pairs are randomly selected.
- Differences are measured on at least an ordinal scale.

- Differences are independent.
- The distribution of differences is approximately symmetric.
- It is used to test whether or not there is a significant difference between two population means when the distribution of the difference between the two samples cannot be assumed to be normal.
- It is used to compare two related samples, matched samples, or to conduct a paired difference test of repeated measurements.

28. Explain Mean with a suitable example. (any correct answer give 5 marks)

Mean (Arithmetic Mean) is the average of a set of observations.
Formula:
Mean = Ex/N
Where:
Ex = Sum of observations
N = Number of observations
Example: Marks of five students = 60, 70, 80, 90, 100
Mean = (60 + 70 + 80 + 90 + 100) / 5
= 400 / 5
= 80
Therefore, the mean mark is 80.